Statistical Report

Determinants of Belgian Household Energy Consumption

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Contents

1 Background **2**

2 Data management **2**

2.1 Data Cleaning 2

2.2 Data Coding 3

3 Descriptive Statistics **3**

3.1 Categorical variables 3

3.2 Continuous variables 3

4 Inferential Statistics **4**

4.1 Regression model 4

4.2 Diagnostic plots 10

4.3 Contrast analysis 10

5 Conclusions **10**

6 Appendix **10**

6.1 Summary tables for non-continuous varibales 10

6.2 Summary tables for continuous varibales 13

6.3 Summary plots for continuous varibales 14

References **20**

1 Background

The goal of "*Carbon Neutrality*" is gaining increasing momentum in the European political arena amid substantial indicators of an incoming climate change which might have deleterious effects at the societal, economic and, most importantly, environmental level. Household energy consumption plays a pivotal role in overall social sustainability since it accounts for a substantial share of the total energy demand in most developed countries. Within this sector, electricity consumption provides a useful proxy for household energy demand and can be put into relationship to structural characteristics of the buildings as well as the socio-economic profile of their occupants.

Dwelling-related attributes such as floor area, dwelling type, construction period, and Energy Performance Certificate (EPC) labels are consistently identified as important predictors of electricity demand (McLoughlin et al., 2013; Santin et al., 2009). Intuitively, larger dwellings typically require more energy for lighting, heating, cooling, and appliance use, while older buildings are constructed with materials which provides poorer insulation and therefore larger heat dissipation. Nevertheless, these variables interacts with socio-economic factors. Larger households topically host a higher number of occupants, therefore they can share the consumption for heating, refrigerators and air conditioning lowering the *per capita* consumption. Similarly, high EPC label buildings are prerogative of high-income classes which might use a larger number of appliances and indulge into energy-consuming behaviours going to erode the benefits of a properly insulated house in terms of energy savings (Aydin et al., 2017).

Using household electricity consumption as a proxy for energy demand, this cross-sectional study examines its relationship with building characteristics (e.g. floor area, EPC label, construction era) and socio-economic factors (number of permanent residents and annual net household income) collecting data from 2500 households in Belgium. Importantly, the study aims at characterising the interplay between EPC labels and the other factors involved in energy consumptions to provide data-driven tools to decision-makers.

2 Data management

2.1 Data Cleaning

After loading the data, three observations (rows 66, 36, and 135) were removed from the dataset according to the instructions. Subsequently, the dataset was cleaned by excluding observations with missing values in the variables EPC label (`epc_label`) and annual household electricity consumption (`annual_kwh`). This step ensures that the analysis is conducted only on complete cases for the key variables of interest. None of the variables displayed missingness above 5%. Missing values in other variables were retained (see appendix) without imputation. To inspect the distribution of electricity consumption and identify potential extreme values, the observations were sorted in decreasing order of annual electricity consumption (`annual_kwh`). Five observations (row 267, 1330, 1570, 369, 12) were two orders of magnitude larger than the others (172 to 312 times larger) and, therefore, were removed as were attributed to imputation error.

2.2 Data Coding

Variables were coded according to their type: continuous (e.g., `annual_kwh`, `sq_meters`, `income_euro`), count (`occupancy_count`), ordinal (`epc_label`, `building_era`), categorical (`roof_color`), or binary (`pet_ownership`). Table 1 summarizes their descriptions and coding schemes.

Table 1: Data Dictionary

Name	Description	Type_Coding
annual_kwh	yearly electricity use	Continuous (kWh/year)
sq_meters	indoor living area	Continuous (m^2)
epc_label	energy performance certificate	Ordinal: A, B, ..., F
building_era	construction period	Ordinal: dummy coded; ref = Pre-1970
income_euro	net household income	Continuous (€/year)
occupancy_count	permanent residents	Count: 1, 2, ..., n
dist_to_brussels	distance to Brussels center	Continuous (km)
roof_color	roof tile color	Categorical: e.g., red, black, other
pet_ownership	owner has a pet	Binary: 0 = No, 1 = Yes

3 Descriptive Statistics

3.1 Categorical variables

The distribution of EPC labels is relatively balanced across categories. Labels A (22.2%) and C (21.3%) are the most common, followed by B (17.6%) and D (15.7%). Lower-efficiency categories E occur less frequently, 13.6% and F 9.6% respectively. Buildings constructed between 1970 and 2000 represent 34.4% of the sample, pre-1970 buildings account for 33.1%, and post-2000 buildings make up 32.6%. Occupancy count ranges from 1 to 7 with all counts accounting approximately 14-15%. Red roofs are the most common (35.2%), followed closely by black roofs (33.3%) and other colours (31.5%). Almost one in four (39.6%) owners has a pet, while 60.4% does not. None of the categories appear too sparse, suggesting that all variables can be retained for modelling without concerns about low cell counts. Overall, the categorical variables show meaningful variation across categories. From the exploratory analysis, two categorical variables seem to correlate strongly with electricity consumption: EPC label and occupancy count. The other categorical variables do njot display an apparent correlation with electricity consumption. Summary tables and plots are presented in the appendix.

3.2 Continuous variables

On average, households consume 5,874.76 kWh of electricity annually, with a relatively large standard deviation (2,224.25 kWh). The median consumption (5,432.07 kWh) is slightly lower than the mean, suggesting a right-skewed distribution, which is also supported by the QQ plot showing upward deviations above the diagonal line in the upper tail (see appendix). Floor area averages

158.25 m² with a standard deviation of 36.50 m², and a range from 60.26 m² to 341.30 m². The QQ plot for floor area shows only mild departures from normality. The average distance to Brussels is 62.96 km, with observations ranging from 5.11 km to 120 km. Its QQ plot indicates noticeable curvature, implying that the distribution deviates from normality, possibly reflecting clustering of households at certain distance ranges during the sampling procedure. Household income averages €48,842.90, with a standard deviation of €22,534.97 and a range from €8,487.91 to €206,367.91. As expected for this type of variable the QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers.

The correlation matrix reveals that annual electricity consumption has a moderate positive correlation with floor area (r = 0.38) and with income (r = 0.29), suggesting that larger and wealthier households tend to consume more energy. Importantly, two predictors exhibit a strong positive correlation with each other, namely floor area and income (r = 0.72). Distance to Brussels shows virtually no correlation with the other variables, suggesting that location relative to Brussels does not strongly influence household energy use, dwelling size, or income in this dataset.

Summary tables and graphs are provided in the appendix.

4 Inferential Statistics

4.1 Regression model

```
## Start:  AIC=33326.27
## annual_kwh ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + roof_color + pet_ownership +
##     income_euro:occupancy_count + epc_label:sq_meters + epc_label:occupancy_count +
##     income_euro:epc_label
##
##                                Df Sum of Sq        RSS   AIC
## - epc_label:income_euro         5   2336998 1567509965 33291
## - sq_meters:epc_label           5   8864937 1574037904 33301
## - roof_color                    2     43385 1565216352 33311
## - building_era                  2    584055 1565757022 33312
## - pet_ownership                 1    364618 1565537585 33319
## - dist_to_brussels              1   1216644 1566389612 33320
## <none>                                      1565172967 33326
## - income_euro:occupancy_count  1  73002999 1638175966 33431
## - epc_label:occupancy_count    5 137321118 1702494086 33495
##
## Step:  AIC=33290.88
## annual_kwh ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + roof_color + pet_ownership +
##     income_euro:occupancy_count + sq_meters:epc_label + epc_label:occupancy_count
##
##                                Df Sum of Sq        RSS   AIC
## - roof_color                    2     68507 1567578472 33275
## - building_era                  2    695843 1568205808 33276
## - pet_ownership                 1    329229 1567839194 33284
```

```
## - dist_to_brussels             1   1306499 1568816464 33285
## - sq_meters:epc_label          5  23435859 1590945825 33289
## <none>                                     1567509965 33291
## - income_euro:occupancy_count  1  72684244 1640194210 33395
## - epc_label:occupancy_count    5 137891854 1705401819 33461
##
## Step:  AIC=33275.36
## annual_kwh ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + pet_ownership + income_euro:occupancy_count +
##     sq_meters:epc_label + epc_label:occupancy_count
##
##                                Df Sum of Sq        RSS   AIC
## - building_era                  2    696695 1568275168 33261
## - pet_ownership                 1    340213 1567918686 33268
## - dist_to_brussels              1   1325886 1568904358 33270
## - sq_meters:epc_label           5  23438144 1591016616 33273
## <none>                                     1567578472 33275
## - income_euro:occupancy_count  1  72649424 1640227896 33380
## - epc_label:occupancy_count    5 138094999 1705673471 33445
##
## Step:  AIC=33260.82
## annual_kwh ~ sq_meters + epc_label + income_euro + occupancy_count +
##     dist_to_brussels + pet_ownership + income_euro:occupancy_count +
##     sq_meters:epc_label + epc_label:occupancy_count
##
##                                Df Sum of Sq        RSS   AIC
## - pet_ownership                 1    354137 1568629305 33254
## - dist_to_brussels              1   1295172 1569570340 33255
## - sq_meters:epc_label           5  23591279 1591866446 33259
## <none>                                     1568275168 33261
## - income_euro:occupancy_count  1  72412926 1640688094 33365
## - epc_label:occupancy_count    5 138229703 1706504870 33431
##
## Step:  AIC=33253.57
## annual_kwh ~ sq_meters + epc_label + income_euro + occupancy_count +
##     dist_to_brussels + income_euro:occupancy_count + sq_meters:epc_label +
##     epc_label:occupancy_count
##
##                                Df Sum of Sq        RSS   AIC
## - dist_to_brussels              1   1291596 1569920901 33248
## - sq_meters:epc_label           5  23594864 1592224169 33251
## <none>                                     1568629305 33254
## - income_euro:occupancy_count  1  72612560 1641241865 33358
## - epc_label:occupancy_count    5 138285993 1706915298 33424
##
## Step:  AIC=33247.79
## annual_kwh ~ sq_meters + epc_label + income_euro + occupancy_count +
##     income_euro:occupancy_count + sq_meters:epc_label + epc_label:occupancy_count
```

```
##
##                                Df Sum of Sq        RSS   AIC
## - sq_meters:epc_label           5  23203758 1593124659 33245
## <none>                                     1569920901 33248
## - income_euro:occupancy_count  1  72953219 1642874120 33352
## - epc_label:occupancy_count    5 138723903 1708644804 33418
##
## Step:  AIC=33245.05
## annual_kwh ~ sq_meters + epc_label + income_euro + occupancy_count +
##     income_euro:occupancy_count + epc_label:occupancy_count
##
##                                Df Sum of Sq        RSS   AIC
## <none>                                     1593124659 33245
## - income_euro:occupancy_count  1  70865951 1663990610 33345
## - epc_label:occupancy_count    5 137412569 1730537229 33411
## - sq_meters                    1 772750486 2365875145 34217
```

```
##
## Call:
## lm(formula = annual_kwh ~ sq_meters + epc_label + income_euro +
##     occupancy_count + income_euro:occupancy_count + epc_label:occupancy_count,
##     data = df_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2848.0  -531.3   -18.2   463.4  3660.8
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  -8.723e+02  1.343e+02  -6.495 1.00e-10 ***
## sq_meters                     2.229e+01  6.450e-01  34.557  < 2e-16 ***
## epc_labelB                    1.286e+02  1.171e+02   1.098 0.272154
## epc_labelC                    4.153e+02  1.108e+02   3.750 0.000181 ***
## epc_labelD                    4.531e+02  1.235e+02   3.669 0.000249 ***
## epc_labelE                    7.014e+02  1.284e+02   5.463 5.15e-08 ***
## epc_labelF                    1.030e+03  1.366e+02   7.537 6.71e-14 ***
## income_euro                  -1.326e-02  1.820e-03  -7.283 4.37e-13 ***
## occupancy_count               5.355e+02  2.512e+01  21.322  < 2e-16 ***
## income_euro:occupancy_count  3.909e-03  3.735e-04  10.465  < 2e-16 ***
## epc_labelB:occupancy_count   8.445e+01  2.599e+01   3.249 0.001174 **
## epc_labelC:occupancy_count   1.096e+02  2.482e+01   4.416 1.05e-05 ***
## epc_labelD:occupancy_count   2.342e+02  2.709e+01   8.645  < 2e-16 ***
## epc_labelE:occupancy_count   3.024e+02  2.829e+01  10.689  < 2e-16 ***
## epc_labelF:occupancy_count   3.566e+02  3.216e+01  11.088  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 804.4 on 2462 degrees of freedom
```

```
##   (5 observations deleted due to missingness)
## Multiple R-squared:   0.87,  Adjusted R-squared:  0.8693
## F-statistic:  1177 on 14 and 2462 DF,  p-value: < 2.2e-16
```

```
## [1] 40282.25
```

```
## Start:  AIC=-10179.75
## log(annual_kwh) ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + roof_color + pet_ownership +
##     income_euro:occupancy_count + epc_label:sq_meters + epc_label:occupancy_count +
##     income_euro:epc_label
##
##                                Df Sum of Sq    RSS    AIC
## - epc_label:income_euro         5  0.034364 36.900 -10216
## - epc_label:occupancy_count     5  0.073973 36.939 -10214
## - sq_meters:epc_label           5  0.089943 36.955 -10213
## - roof_color                    2  0.007997 36.873 -10195
## - building_era                  2  0.008538 36.874 -10195
## - income_euro:occupancy_count   1  0.002529 36.868 -10187
## - pet_ownership                 1  0.013821 36.879 -10187
## - dist_to_brussels              1  0.030355 36.896 -10186
## <none>                                      36.865 -10180
##
## Step:  AIC=-10216.52
## log(annual_kwh) ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + roof_color + pet_ownership +
##     income_euro:occupancy_count + sq_meters:epc_label + epc_label:occupancy_count
##
##                                Df Sum of Sq    RSS    AIC
## - epc_label:occupancy_count     5  0.071900 36.971 -10251
## - sq_meters:epc_label           5  0.177529 37.077 -10244
## - roof_color                    2  0.008645 36.908 -10232
## - building_era                  2  0.009863 36.909 -10232
## - income_euro:occupancy_count   1  0.002580 36.902 -10224
## - pet_ownership                 1  0.012962 36.912 -10224
## - dist_to_brussels              1  0.031456 36.931 -10222
## <none>                                      36.900 -10216
##
## Step:  AIC=-10250.78
## log(annual_kwh) ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + roof_color + pet_ownership +
##     income_euro:occupancy_count + sq_meters:epc_label
##
##                                Df Sum of Sq    RSS    AIC
## - sq_meters:epc_label           5  0.179565 37.151 -10278
## - roof_color                    2  0.007796 36.979 -10266
## - building_era                  2  0.010650 36.982 -10266
```

```
## - income_euro:occupancy_count  1  0.002785 36.974 -10258
## - pet_ownership                1  0.013668 36.985 -10258
## - dist_to_brussels             1  0.033084 37.005 -10256
## <none>                                     36.971 -10251
##
## Step:  AIC=-10277.87
## log(annual_kwh) ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + roof_color + pet_ownership +
##     income_euro:occupancy_count
##
##                                Df Sum of Sq    RSS      AIC
## - roof_color                    2    0.0076 37.159 -10293.0
## - building_era                  2    0.0121 37.163 -10292.7
## - income_euro:occupancy_count  1    0.0059 37.157 -10285.3
## - pet_ownership                1    0.0106 37.162 -10285.0
## - dist_to_brussels             1    0.0299 37.181 -10283.7
## <none>                                     37.151 -10277.9
## - epc_label                     5    8.3530 45.504  -9814.6
## - sq_meters                     1   23.8999 61.051  -9055.3
##
## Step:  AIC=-10292.99
## log(annual_kwh) ~ sq_meters + epc_label + building_era + income_euro +
##     occupancy_count + dist_to_brussels + pet_ownership + income_euro:occupancy_count
##
##                                Df Sum of Sq    RSS      AIC
## - building_era                  2    0.0119 37.171 -10307.8
## - income_euro:occupancy_count  1    0.0060 37.165 -10300.4
## - pet_ownership                1    0.0108 37.169 -10300.1
## - dist_to_brussels             1    0.0295 37.188 -10298.8
## <none>                                     37.159 -10293.0
## - epc_label                     5    8.3757 45.534  -9828.6
## - sq_meters                     1   23.9261 61.085  -9069.6
##
## Step:  AIC=-10307.83
## log(annual_kwh) ~ sq_meters + epc_label + income_euro + occupancy_count +
##     dist_to_brussels + pet_ownership + income_euro:occupancy_count
##
##                                Df Sum of Sq    RSS      AIC
## - income_euro:occupancy_count  1     0.006 37.177 -10315.2
## - pet_ownership                1     0.011 37.182 -10314.9
## - dist_to_brussels             1     0.029 37.199 -10313.7
## <none>                                     37.171 -10307.8
## - sq_meters                     1    23.916 61.087  -9085.1
## - epc_label                     5    41.883 79.053  -8477.8
##
## Step:  AIC=-10315.23
## log(annual_kwh) ~ sq_meters + epc_label + income_euro + occupancy_count +
##     dist_to_brussels + pet_ownership
```

```
##
##                     Df Sum of Sq     RSS      AIC
## - income_euro        1     0.006  37.182 -10322.7
## - pet_ownership      1     0.011  37.188 -10322.3
## - dist_to_brussels   1     0.028  37.205 -10321.2
## <none>                            37.177 -10315.2
## - sq_meters          1    23.915  61.092  -9092.7
## - epc_label          5    41.876  79.053  -8485.6
## - occupancy_count    1   219.846 257.023  -5533.8
##
## Step:  AIC=-10322.67
## log(annual_kwh) ~ sq_meters + epc_label + occupancy_count + dist_to_brussels +
##     pet_ownership
##
##                     Df Sum of Sq     RSS      AIC
## - pet_ownership      1     0.011  37.194 -10329.7
## - dist_to_brussels   1     0.029  37.211 -10328.6
## <none>                            37.182 -10322.7
## - epc_label          5    41.876  79.058  -8493.2
## - sq_meters          1    51.556  88.738  -8175.9
## - occupancy_count    1   219.953 257.136  -5540.5
##
## Step:  AIC=-10329.75
## log(annual_kwh) ~ sq_meters + epc_label + occupancy_count + dist_to_brussels
##
##                     Df Sum of Sq     RSS      AIC
## - dist_to_brussels   1     0.029  37.223 -10335.6
## <none>                            37.194 -10329.7
## - epc_label          5    41.895  79.089  -8500.1
## - sq_meters          1    51.546  88.739  -8183.7
## - occupancy_count    1   220.130 257.324  -5546.6
##
## Step:  AIC=-10335.63
## log(annual_kwh) ~ sq_meters + epc_label + occupancy_count
##
##                    Df Sum of Sq     RSS      AIC
## <none>                           37.223 -10335.6
## - epc_label         5    41.997  79.219  -8503.8
## - sq_meters         1    51.520  88.743  -8191.4
## - occupancy_count   1   220.102 257.325  -5554.4
```

```
##
## Call:
## lm(formula = log(annual_kwh) ~ sq_meters + epc_label + occupancy_count,
##     data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -0.46693 -0.08015  0.00119  0.08011  0.40719
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     7.210e+00  1.304e-02  552.97   <2e-16 ***
## sq_meters       3.958e-03  6.771e-05   58.46   <2e-16 ***
## epc_labelB      8.743e-02  7.872e-03   11.11   <2e-16 ***
## epc_labelC      1.592e-01  7.483e-03   21.27   <2e-16 ***
## epc_labelD      2.418e-01  8.145e-03   29.68   <2e-16 ***
## epc_labelE      3.254e-01  8.501e-03   38.28   <2e-16 ***
## epc_labelF      4.031e-01  9.532e-03   42.29   <2e-16 ***
## occupancy_count 1.511e-01  1.251e-03  120.83   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1228 on 2469 degrees of freedom
##   (5 observations deleted due to missingness)
## Multiple R-squared:  0.8911, Adjusted R-squared:  0.8908
## F-statistic:  2887 on 7 and 2469 DF,  p-value: < 2.2e-16
```

```
## [1] -3298.412
```

4.2 Diagnostic plots

- Residuals vs Fitted
- Normal Q–Q Plot
- Scale–Location Plot (Spread–Location)
- Residuals vs Leverage
- Influence (Cook's Distance)

4.3 Contrast analysis

- for research question 2 and 4

5 Conclusions

6 Appendix

6.1 Summary tables for non-continuous varibales

Table 2: EPC Label Distribution

epc_label	n	percent
A	552	22.2

B	436	17.6
C	529	21.3
D	389	15.7
E	337	13.6
F	239	9.6

Table 3: Building Era Distribution

building_era	n	percent
Pre-1970	821	33.1
1970-2000	853	34.4
Post-2000	808	32.6

Table 4: Roof Color Distribution

roof_color	n	percent
Black	826	33.3
Other	783	31.5
Red	873	35.2

Table 5: Pet Ownership Distribution

pet_ownership	n	percent
0	1500	60.4
1	982	39.6

Table 6: Occupancy Count Distribution

occupancy_count	n	percent
1	350	14.1
2	354	14.3
3	342	13.8
4	376	15.1
5	375	15.1
6	341	13.7
7	339	13.7
NA	5	0.2

6.2 Summary tables for continuous varibales

Table 7: Summary Statistics for Continuous Variables

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	5874.76	158.25	62.96	48842.90
SD	2224.25	36.50	33.04	22534.97
Median	5432.07	155.86	63.21	44535.07
IQR	2987.79	46.95	57.55	25894.30
Min	1824.63	60.26	5.11	8487.91
Max	17197.12	341.30	120.00	206367.91

6.3 Summary plots for continuous varibales

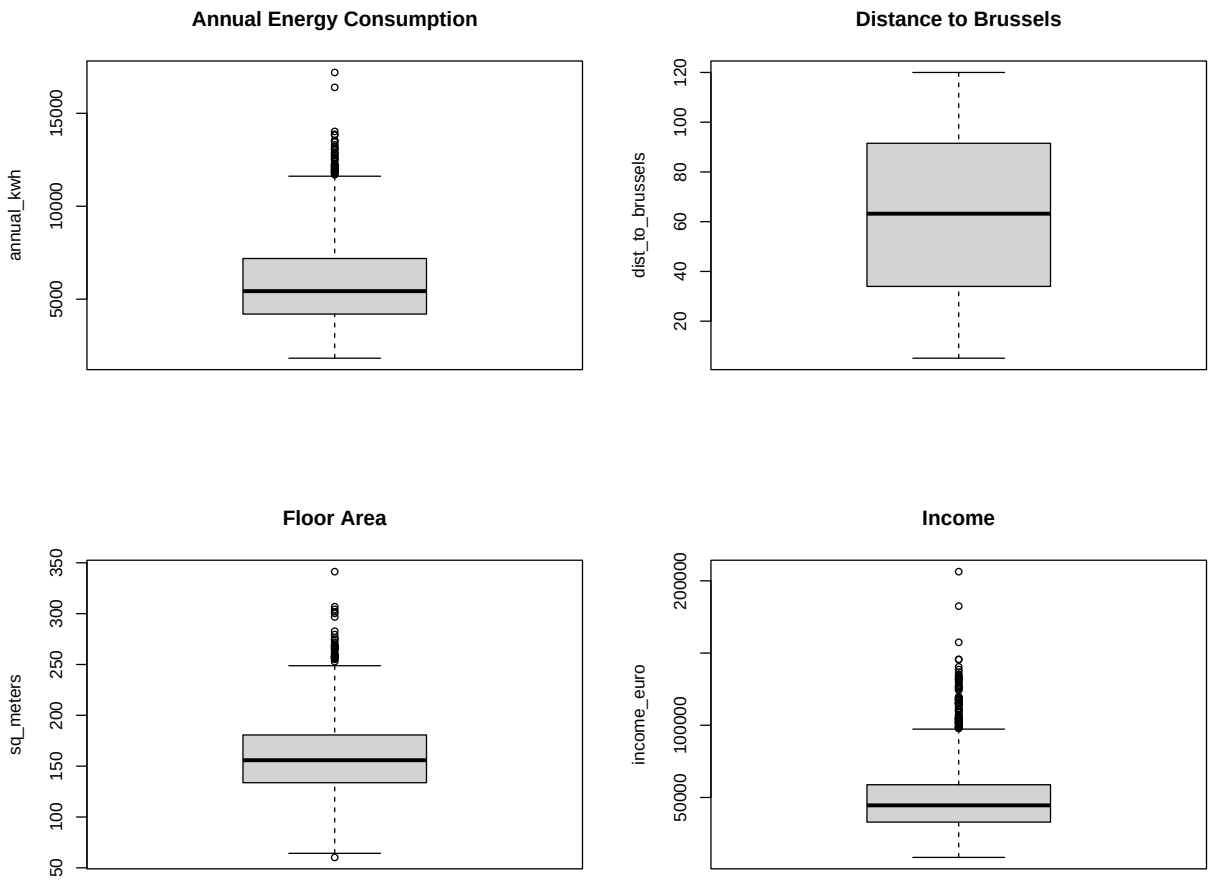


Figure 1: Box plots for continuous variables

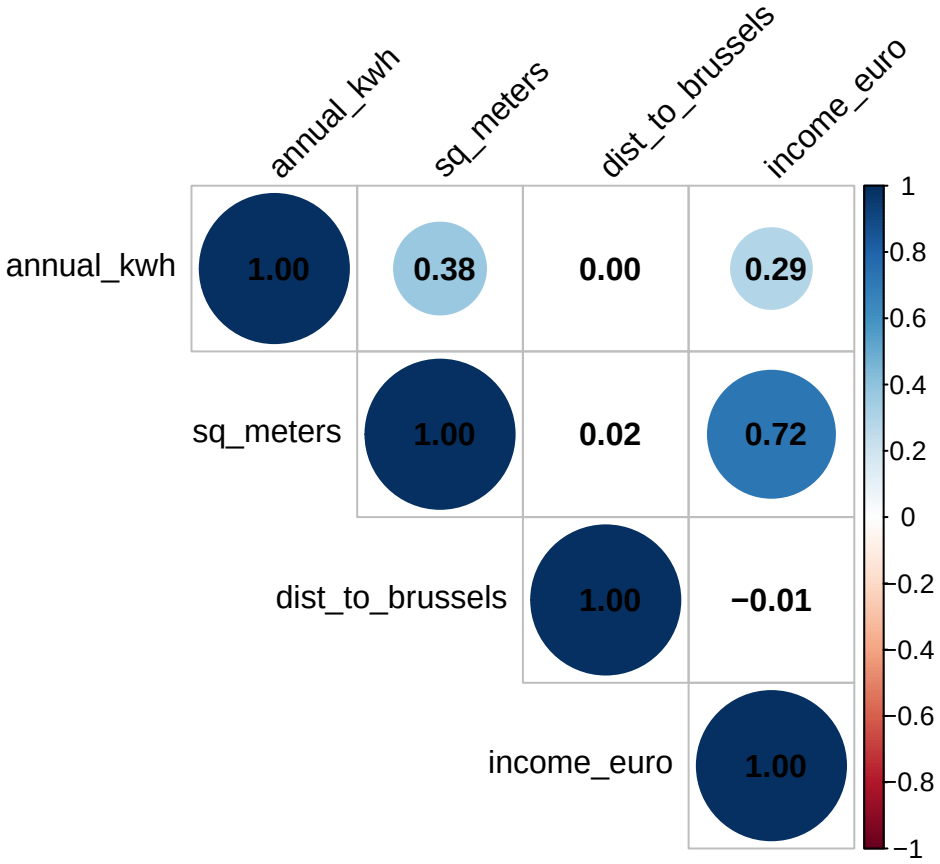


Figure 2: Correlation plots for continuous variables

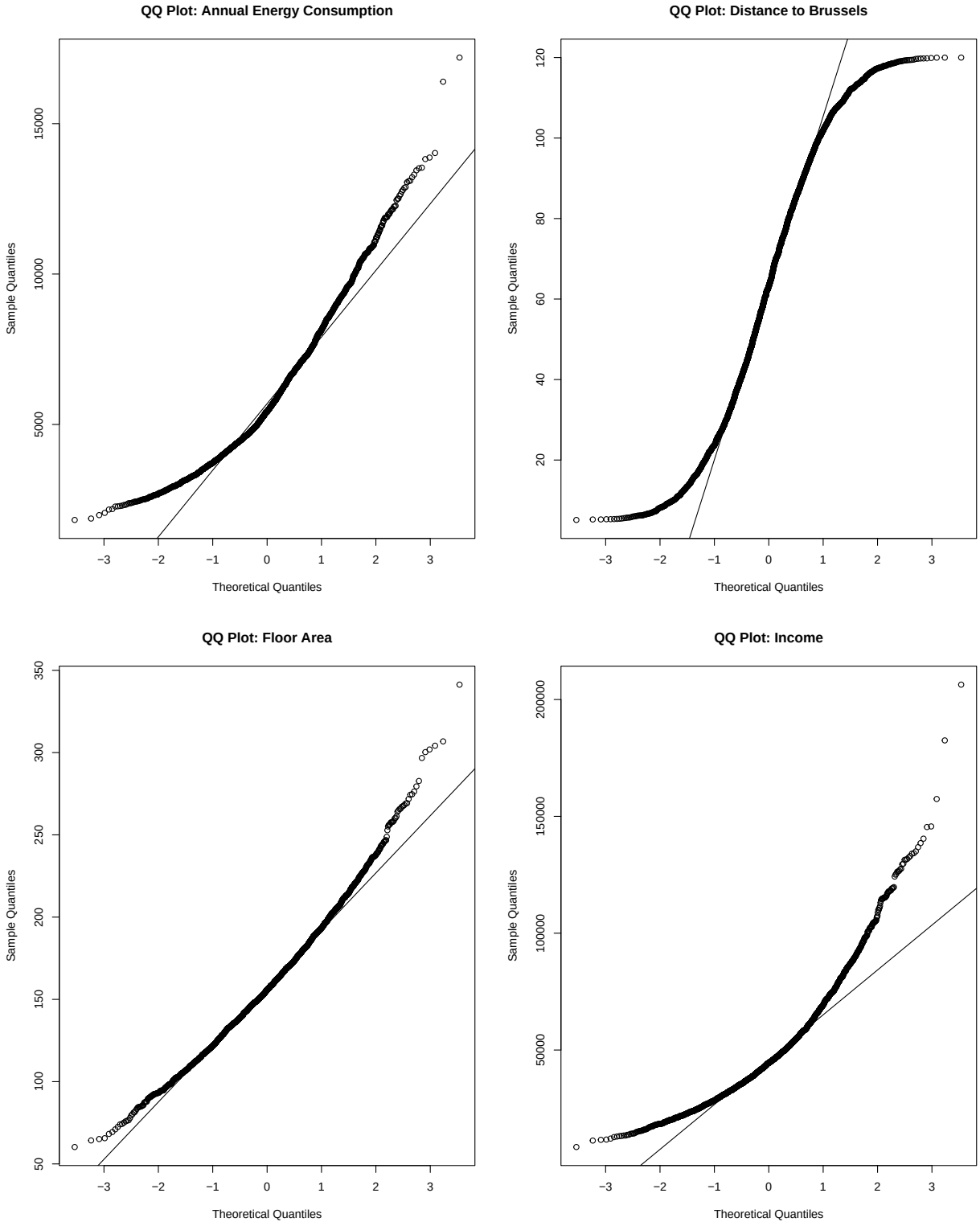


Figure 3: QQ plots for continuous variables

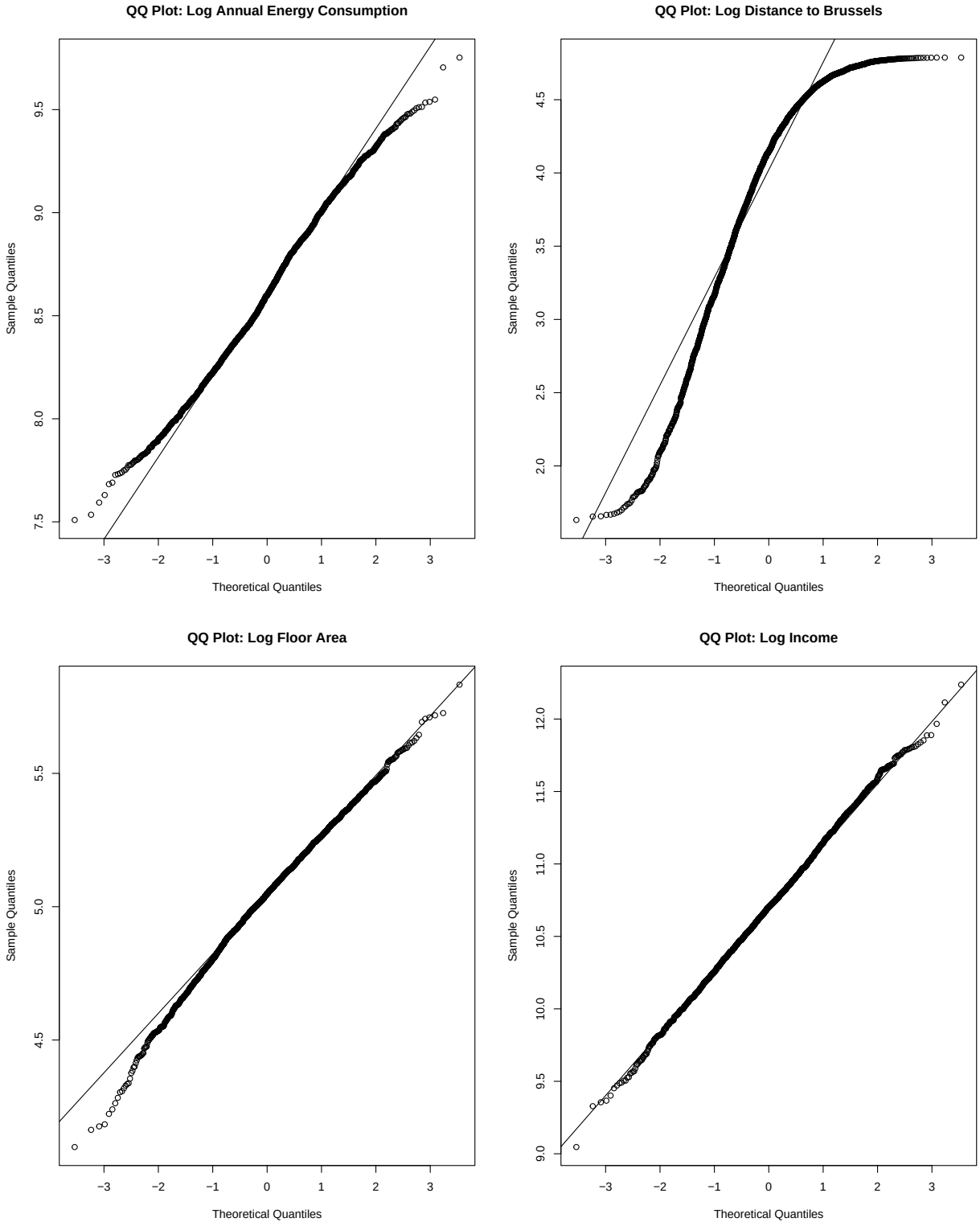
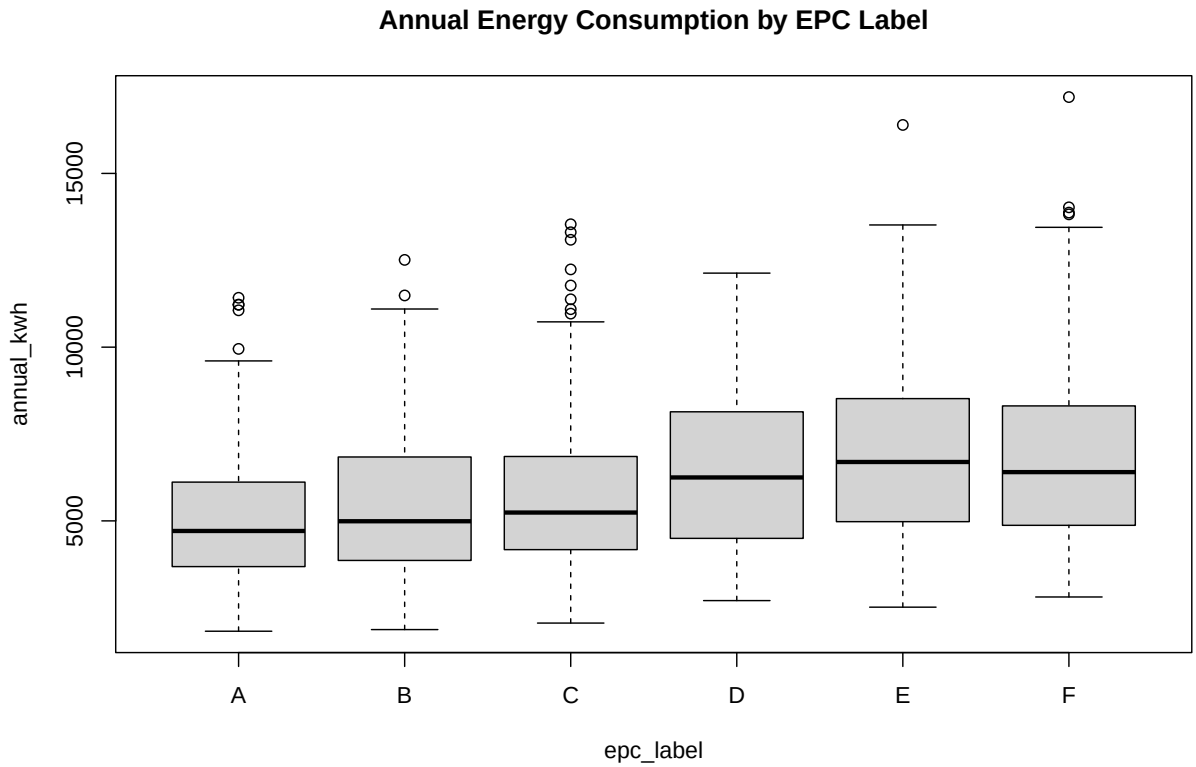


Figure 4: QQ plots for log-transformed continuous variables



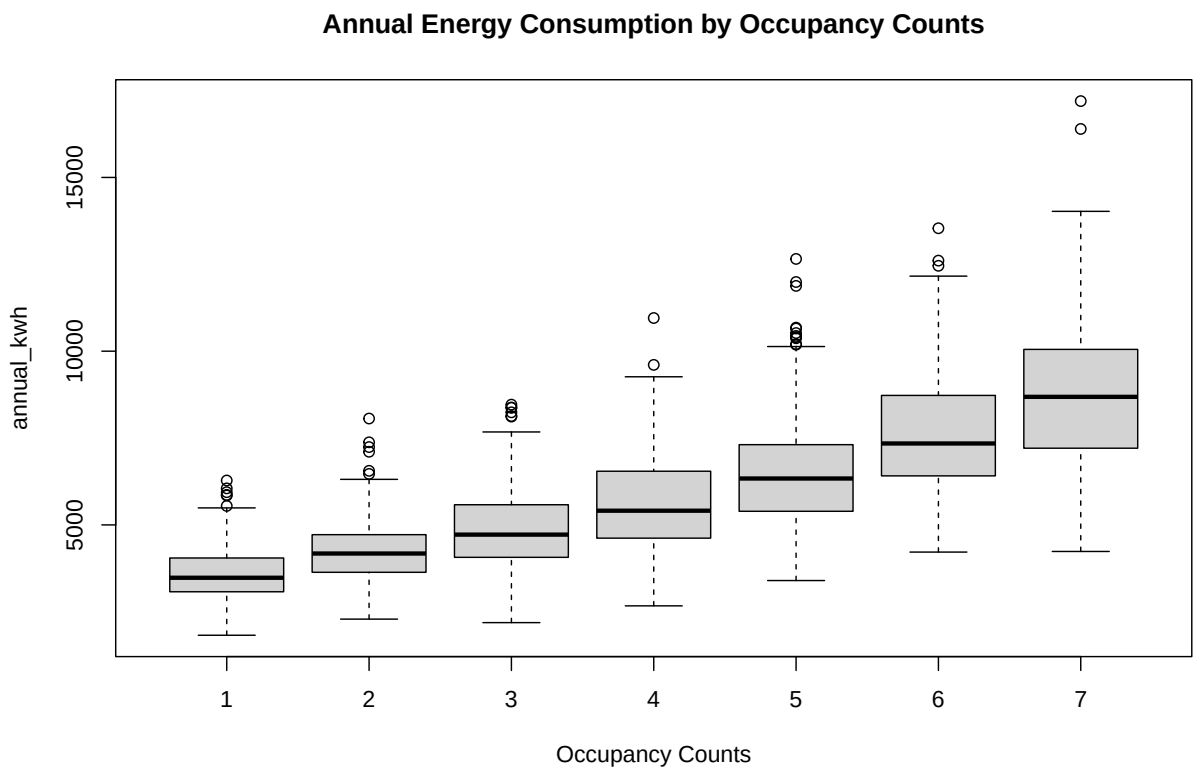


Figure 5: Box plots for annual energy consumption vs categorical variables (part 1)

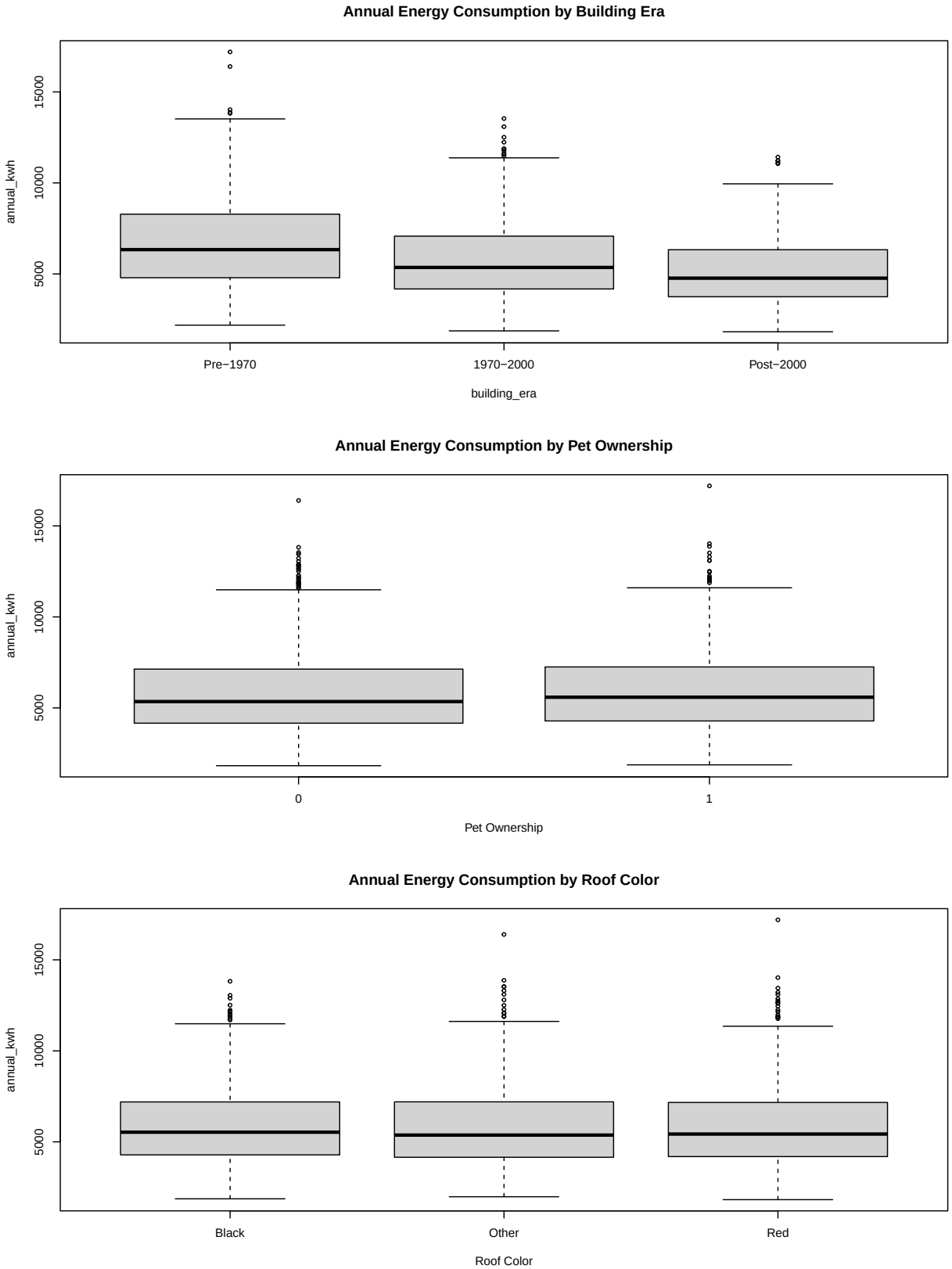


Figure 6: Box plots for annual energy consumption vs categorical variables (part 2)

References

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